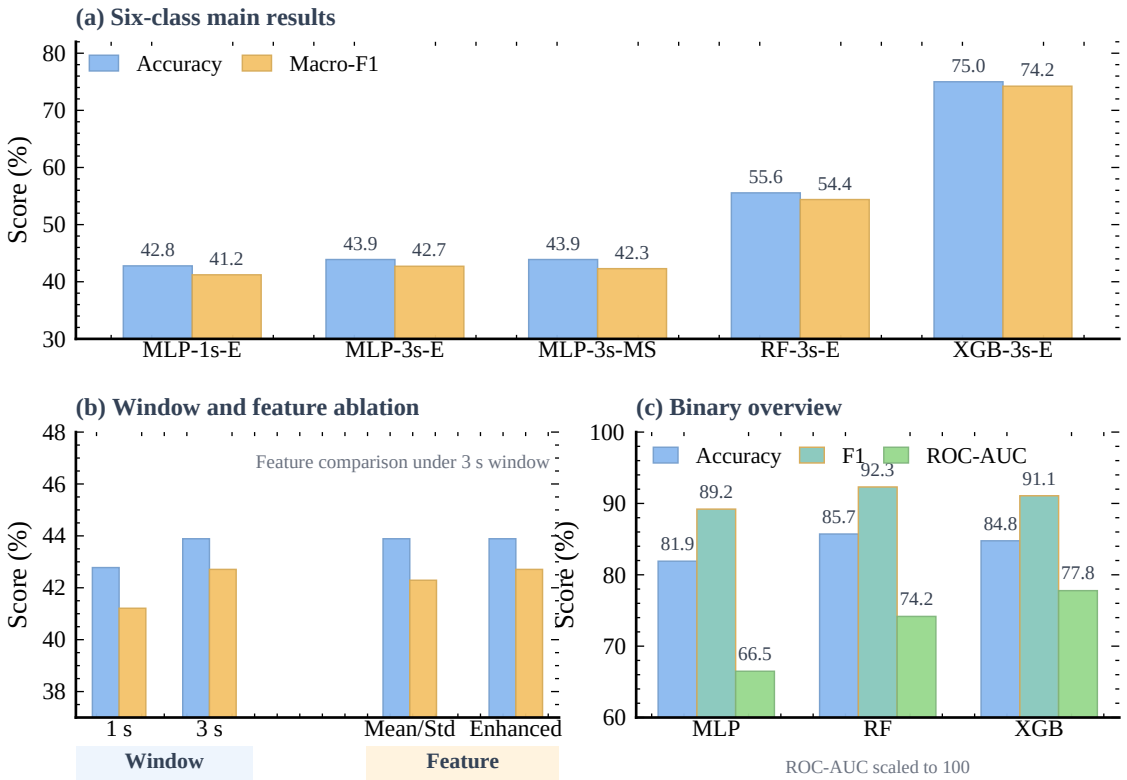
(a) Six-class main results
Accuracy
Macro-F1
Score (%)
80
70
60
50
40
30
42.8
41.2
43.9
42.7
43.9
42.3
55.6
54.4
75.0
74.2
MLP-1s-E
MLP-3s-E
MLP-3s-MS
RF-3s-E
XGB-3s-E
(b) Window and feature ablation
Feature comparison under 3 s window
Score (%)
48
46
44
42
40
38
1 s
3 s
Mean/Std
Enhanced
Window
Feature
(c) Binary overview
Accuracy
F1
ROC-AUC
Score (%)
100
90
80
70
60
81.9
89.2
66.5
85.7
92.3
74.2
84.8
91.1
77.8
MLP
RF
XGB
ROC-AUC scaled to 100